Database

DataConstants

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DataLoader

-getJsonFile(String fileName): File
+getUsers(): ArrayList<User>
+getShelters(): ArrayList<Shelter>
+getReliefRequests(): ArrayList<ReliefRequest>
+getHurricanes(): ArrayList<Hurricane>

DataWriter

+saveUsers(): boolean
+saveShelters(ArrayList<Shelter> shelters): boolean
+saveReliefRequests(ArrayList<ReliefRequest> reliefRequests): boolean
+saveHurricanes(ArrayList<Hurricane> hurricanes): boolean
+getShelterJSON(Shelter shelter): JSONObject
+getReliefRequestJSON(ReliefRequest reliefRequest): JSONObject
+getHurricaneJSON(Hurricane hurricane): JSONObject
+getUserJSON(User user): JSONObject

Facades

User UI

...

...

UserSystem

-userList: UserList
-reliefRequestList: ReliefRequestList
-shelterList: ShelterList
-hurricaneList: HurricaneList
-currentUser: User
-currentHurricane: Hurricane

+login(String userName, String password): User
+logout(): void
+getUser(): User
+signup(String firstName, String lastName, LocalDate birthDate, String userName, String password, String email, Location location): User
+becomeVolunteer(String userName, String password): User
+getUsers(): ArrayList<User>
+getShelters(): ArrayList<Shelter>
+findShelters(ArrayList<Filter> requestedFilters): ArrayList<Shelter>
+getShelterManagers(Shelter shelter): ArrayList<ShelterManager>
+getVolunteers(): ArrayList<Volunteer>
+getReliefRequests(): ArrayList<ReliefRequest>
+getHurricane(): Hurricane
+getHurricane(Location location): Hurricane
+createReliefRequest(String userName, ArrayList<Victim> victims, ArrayList<Hazard> hazards, String photo, ArrayList<AssistanceType> assistanceTypes, Status status, String description, String phoneNumber, UUID id, Location location): ReliefRequest
+submitReliefRequest(ReliefRequest request): void
+editReliefRequest(ReliefRequest request, String description, Status status): void
+assignUserToShelter(Shelter shelter, Volunteer volunteer): boolean
+acceptReliefRequest(ReliefRequest request): void
+acceptRequest(Volunteer volunteer, ReliefRequest request): void
+getCurrentUser(): User
+getUser(String userName, String password): User
+findNearestReliefRequest(Location location): ReliefRequest

Singletons/Lists

HurricaneList

-hurricaneList: HurricaneList
-hurricanes: ArrayList<Hurricane>

+HurricaneList()
+getInstance(): HurricaneList
+addHurricane(String hurricaneName, Location location, int category, double windSpeed, double diameter): void
+updateHurricane(UUID id, Location newLocation, int newCategory, double newWindSpeed, double newDiameter): void
+getHurricane(Location location): Hurricane
+getHurricanes(): ArrayList<Hurricane>
+removeHurricane(UUID id): void
+saveHurricanes(): boolean

ShelterList

-shelterList: ShelterList
-shelters: ArrayList<Shelter>

+ShelterList()
+getInstance(): ShelterList
+getShelter(String shelterName, Location location): Shelter
+findShelters(ArrayList<Filter> requestedFilters): ArrayList<Shelter>
+getShelters(): ArrayList<Shelter>
+addShelter(Shelter shelter): void
+removeShelter(Shelter shelter): void
+saveShelters(): boolean

UserList

-userList: UserList
-users: ArrayList<User>

+UserList()
+getInstance(): UserList
+getUser(String userName, String password): User
+addUser(String firstName, String lastName, LocalDate birthDate, String userName, String password, String email, Location location): boolean
+addVolunteer(Volunteer volunteer): void
+getUsers(): ArrayList<User>
+getVolunteers(): ArrayList<Volunteer>
+getShelterManagers(): ArrayList<User>
+saveUser(): boolean

ReliefRequestList

-reliefRequestList: ReliefRequestList
-reliefRequests: ArrayList<ReliefRequest>

+ReliefRequestList()
+getInstance(): ReliefRequestList
+addReliefRequest(ReliefRequest reliefRequest): void
+getReliefRequests(): ArrayList<ReliefRequest>
+getReliefRequest(String phoneNumber): ReliefRequest
+getActiveReliefRequestById(String id): ReliefRequest
+findNearestReliefRequest(Location location): ReliefRequest
+saveReliefRequests(): boolean

Hurricane

-hurricaneName: String
-location: Location
-category: int
-windSpeed: double
-diameter: double
-id: UUID

+Hurricane(String hurricaneName, Location location, int category, double windSpeed, double diameter)
+setLocation(Location location): void
+setHurricaneName(String hurricaneName): void
+getName(): String
+setCategory(int category): void
+setWindSpeed(double windSpeed): void
+setDiameter(double diameter): void
+getHurricaneName(): String
+getLocation(): Location
+getCategory(): int
+getWindSpeed(): double
+getDiameter(): double
+getId(): UUID

<<enumeration>> Filter

PET_FRIENDLY
FOOD
WATER
MEDICAL
OPEN_CAPACITY

Shelter

Shelter

-shelterName: String
-filters: ArrayList<Filter>
-shelterManagers: ArrayList<ShelterManager>
-capacity: int
-occupancy: int
-location: Location
-availability: Availability
-id: UUID

+Shelter(String shelterName, int capacity, Location location)
+getShelterName(): String
+setShelterName(String shelterName): void
+getFilters(): ArrayList<Filter>
+addFilter(Filter filter): void
+getShelterManagers(): ArrayList<ShelterManager>
+addShelterManager(ShelterManager shelterManager): void
+getCapacity(): int
+setCurrentCapacity(int currentCapacity): void
+setCapacity(int capacity): void
+setCurrentOccupancy(int currentOccupancy): void
+getOccupancy(): int
+getAvailability(): Availability
+getLocation(): Location
+getId(): UUID

ShelterManager

-shelterVolunteers: ArrayList<Volunteer>

+ShelterManager(String firstName, String lastName, LocalDate birthDate, String email, String password, String username, Location location)
+addShelterVolunteer(Volunteer volunteer): void
+removeShelterVolunteer(Volunteer volunteer): void
+getShelterVolunteers(): ArrayList<Volunteer>
+updateShelterCapacity(Shelter shelter, int capacity): void

User

-firstName: String
-lastName: String
-birthDate: LocalDate
-email: String
-password: String
-userName: String
-location: Location
-id: UUID
-requests: ArrayList<ReliefRequest>

+User(String firstName, String lastName, LocalDate birthDate, String email, String password, String username, Location location)
+isMatch(String userName, String password): boolean
+getLocation(): Location
+getFirstName(): String
+getLastName(): String
+getBirthDate(): LocalDate
+getEmail(): String
+getPassword(): String
+getUsername(): String
+getId(): String

ReliefRequest

-userName: String
-victims: ArrayList<Victim>
-hazards: ArrayList<Hazard>
-photo: String
-assistanceTypes: ArrayList<AssistanceType>
-status: Status
-description: String
-phoneNumber: String
-id: UUID
-location: Location
-isAccepted: boolean

+ReliefRequest(String userName, ArrayList<Victim> victims, ArrayList<Hazard> hazards, String photo, ArrayList<AssistanceType> assistanceTypes, Status status, String description, String phoneNumber, UUID id, Location location)
+addVictim(Victim victim): void
+addHazard(Hazard hazard): void
+setPhoto(String photo): void
+addAssistanceType(AssistanceType assistanceType): void
+setStatus(Status status): void
+setDescription(String description): void
+getStatus(): Status
+getDescription(): String
+getPhoneNumber(): String
+getId(): UUID
+getLocation(): Location
+getVictims(): ArrayList<Victim>
+setAccepted(boolean accepted): void
+isAccepted(): boolean
+getUserName(): String
+getHazards(): ArrayList<Hazard>
+getPhoto(): String
+getAssistanceTypes(): ArrayList<AssistanceType>
+getAllDetails(ReliefRequest request): boolean

<<enumeration>> Status

AWAITING_ASSISTANCE
IN_PROGRESS
COMPLETED

<<enumeration>> Assistance Type

MEDICAL
TRANSPORTATION
VETINARY
DEBRIS_REMOVAL

<<enumeration>> Hazard

FLOODING
FIRE
DOWNED_POWER_LINES
GAS_LEAK
DEBRIS
STRUCTURAL_DAMAGE
ROAD_BLOCKAGE

Location

-street: String
-city: String
-state: String
-zipCode: int

+Location(String street, String city, String state, int zipCode)
+getZipCode(): int
+getStreet(): String
+getCity(): String
+getState(): String

Volunteer

Volunteer

-tools ArrayList<tool>
-vehicles ArrayList<vehicle>
-certifications ArrayList<Certification>

+Volunteer(String firstName, String lastName, LocalDate birthDate, String email, String password, String username, Location location)
+acceptRequest(ReliefRequest request): ReliefRequest
+getTools(): ArrayList<Tools>
+getVehicles(): ArrayList<Vehicle>
+getCertifications(): ArrayList<Certification>
+addTool(Tools tool): void
+addVehicle(Vehicle vehicle): void
+addCertification(Certification certification): void

<<enumeration>> Tools

CHAINSAW
SAW
HAMMER
WRENCH
FLASHLIGHT
AXE
RADIO

<<enumeration>> Vehicle

TRUCK
CAR
BOAT
AIR_SUPPORT

<<enumeration>> Certification

CPR
AED
PFA
CERT
NRP

Victim

Victim

-firstName: String
-lastName: String
-description: String
-gender: String
-age: int
-victimStatus: VictimStatus

+Victim(String firstName, VictimStatus victimStatus, int age)
+getFirstName(): String
+getLastName(): String
+getDescription(): String
+getGender(): String
+getAge(): int
+getVictimStatus(): VictimStatus
+setVictimStatus(VictimStatus status): void

<<enumeration>> VictimStatus

LOST
MINOR_INJURY
MAJOR_INJURY
TRAPPED
UNRESPONSIVE
ALRIGHT
DECEASED

Pet

-species: String

+Pet(String name, VictimStatus victimStatus, String species, int age)
+getSpecies(): String